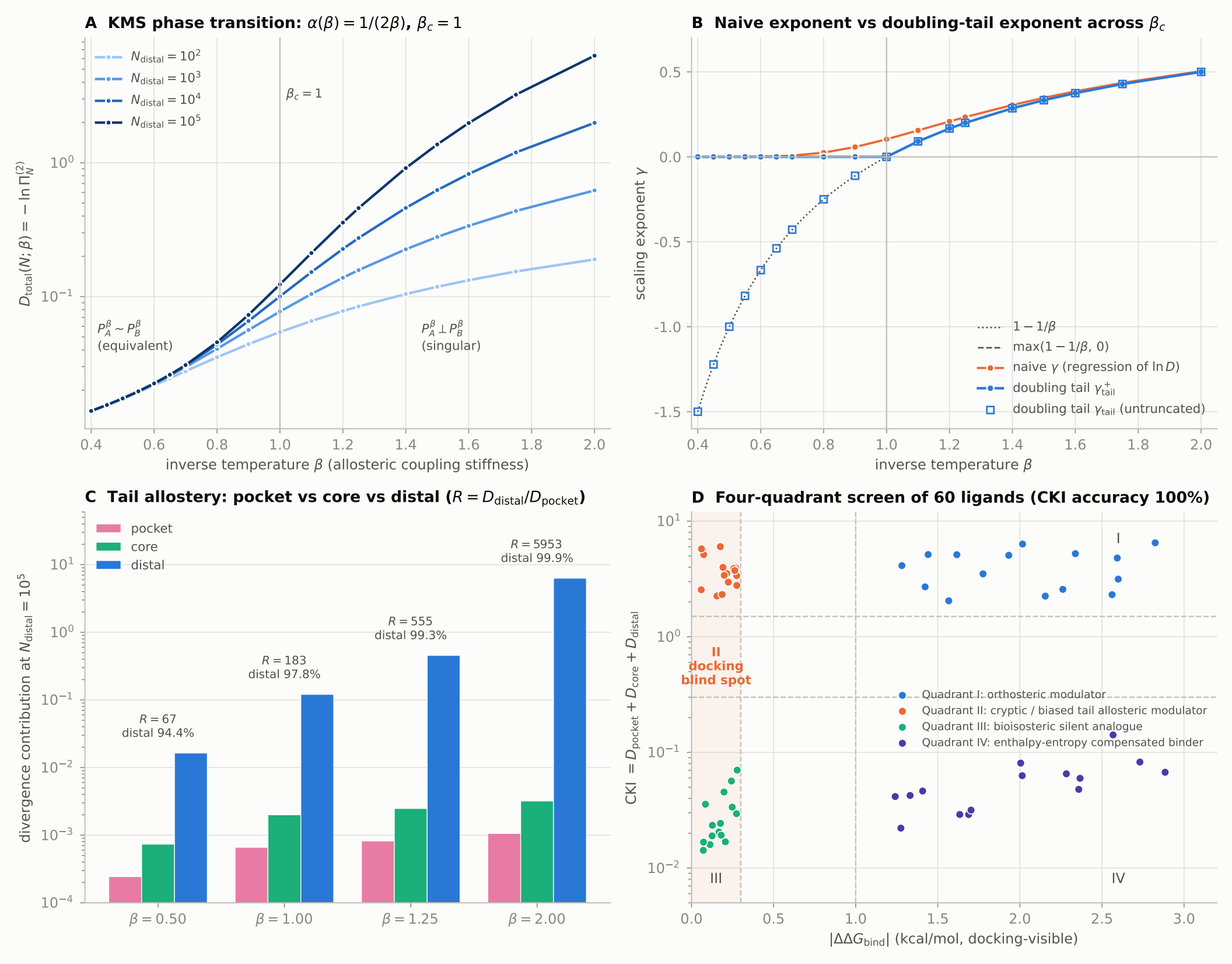
A KMS phase transition: $\alpha(\beta) = 1/(2\beta)$, $\beta_c = 1$
$N_{\text{distal}} = 10^2$
$N_{\text{distal}} = 10^3$
$N_{\text{distal}} = 10^4$
$N_{\text{distal}} = 10^5$
$\beta_c = 1$
$P_A^\beta \sim P_B^\beta$ (equivalent)
$P_A^\beta \perp P_B^\beta$ (singular)
$D_{\text{total}}(N;\beta) = -\ln \Pi_N^{(2)}$
inverse temperature β (allosteric coupling stiffness)
B Naive exponent vs doubling-tail exponent across β_c
scaling exponent γ
$1 - 1/\beta$
$\max(1 - 1/\beta, 0)$
naive γ (regression of $\ln D$)
doubling tail γ_{tail}^+
doubling tail γ_{tail} (untruncated)
inverse temperature β
C Tail allostery: pocket vs core vs distal ($R = D_{\text{distal}}/D_{\text{pocket}}$)
pocket
core
distal
$R = 67$
distal 94.4%
$R = 183$
distal 97.8%
$R = 555$
distal 99.3%
$R = 5953$
distal 99.9%
divergence contribution at $N_{\text{distal}} = 10^5$
$\beta = 0.50$
$\beta = 1.00$
$\beta = 1.25$
$\beta = 2.00$
D Four-quadrant screen of 60 ligands (CKI accuracy 100%)
I
II
docking
blind spot
III
IV
Quadrant I: orthosteric modulator
Quadrant II: cryptic / biased tail allosteric modulator
Quadrant III: bioisosteric silent analogue
Quadrant IV: enthalpy-entropy compensated binder
CKI $= D_{\text{pocket}} + D_{\text{core}} + D_{\text{distal}}$
$|\Delta\Delta G_{\text{bind}}|$ (kcal/mol, docking-visible)